

# Nexora Knee AI

## AI-Assisted Medial Meniscus Assessment & Patient-Specific Knee Implant Sizing - PS Solution Document

6-Hour Hackathon Prototype | CPU-only | Pretrained Deep Learning + Rule-Based Geometry

Stack: Python 3.11 - TensorFlow-CPU 2.15 - Streamlit - Plotly

Status: environment ready, pretrained model downloaded, inference verified on real data

## 1. Problem Statement Summary

Build an intelligent imaging system with two modules plus a visualization interface:

- Module 1 - Meniscus Assessment: segment femur, tibia and medial meniscus; measure meniscus thickness (mm) at predefined locations; correlate findings with osteoarthritis (OA) status, age and sex.
- Module 2 - Implant Sizing: segment femur and tibia; extract mediolateral (ML) and anteroposterior (AP) dimensions; match against an implant sizing database; recommend patient-specific component sizes.
- Visualization UI: interactive overlays of anatomy, measurements and recommendations.
- Positioning: decision-SUPPORT tool assisting clinicians - never autonomous diagnosis.

### 1.1 Key Technical Reality Check

- Meniscus is NOT visible on plain X-ray -> Module 1 requires MRI; OA KL-grading uses X-ray.
- Training segmentation networks needs GPU-days -> infeasible in-hackathon; solved via pretrained models + rule-based geometry (Option A).
- Public knee X-ray datasets carry KL grades but no masks -> thickness is demonstrated as a calibrated joint-space proxy on MRI ROIs.

## 2. Proposed Solution - Nexora Knee AI

A four-tab Streamlit decision-support application that runs entirely on CPU:

- Tab 1 OA GRADER (real deep learning): fine-tuned Xception CNN grades uploaded knee X-rays into KL grades 0-4 (Healthy..Severe) with confidence bars and Grad-CAM explainability heatmap.
- Tab 2 MRI EXPLORER (proxy thickness demo): loads 3D knee MRI volumes (.pck), slice navigation, ROI overlay from dataset metadata, ACL-status badges, and pixel->mm joint-space width measurement as the stand-in for meniscus thickness assessment workflow.
- Tab 3 IMPLANT SIZER (rule-based engine): takes femoral/tibial ML+AP dimensions (manually entered or estimated) and matches them against REAL published Zimmer NexGen component dimensions (femoral A-H, tibial 1-10) using nearest-fit ranking with overhang warnings.
- Tab 4 DATA ANALYTICS: dataset composition, KL grade distributions, ACL-label balance - demonstrates the population-level correlation capability of Module 1.

X-ray upload ??> Xception-ft (224x224x3) ??> KL grade 0-4 + probs + Grad-CAM  
 MRI volume ??> slice nav + ROI overlay (metadata) ??> JSW proxy (px->mm)  
 Dimensions ??> nearest-match vs NexGen chart (fem A-H / tib 1-10) ??> ranked sizes

## 3. PS Requirement -> Implementation Map

| PS Requirement                   | Implementation                                                            | Maturity   |
|----------------------------------|---------------------------------------------------------------------------|------------|
| Segment femur / tibia / meniscus | Metadata-driven ROI localization on MRI volumes; documented path to U-Net | Proxy      |
| Meniscus thickness (mm)          | Calibrated joint-space width measurement on MRI slices (px -> mm)         | Proxy demo |
| Correlate with OA status         | Pretrained Xception KL-grading (real inference) + cohort analytics        | Real AI    |
| Femoral/tibial ML & AP dims      | Measurement inputs + ROI-derived estimates                                | Working    |
| Implant sizing database          | Real Zimmer NexGen published dimensions (fem A-H, tib 1-10)               | Real data  |
| Size recommendation              | Euclidean nearest-match ranking + overhang warning                        | Working    |
| Visualization UI                 | Streamlit overlays: heatmaps, ROI boxes, ranked tables, charts            | Working    |

## 4. Repository Architecture

```

D:/Nexora-26/
|-- medtech_ps.pdf           problem statement (input)
|-- requirements.txt         pinned deps (Python 3.11 venv at .venv/)
|-- PS_Solution_Nexora_Knee_AI.pdf this document
|-- models/model_Xception_ft.hdf5 pretrained weights (251MB, verified loading)
|-- data/
|   |-- knee-osteoarthritis-severity/ 9,786 X-rays, KL 0-4, train/val/test
|   |-- vol01..vol08/*.pck           917 knee MRI volumes (32x320x320 uint16)
|   |-- metadata.csv                examId, aclDiagnosis(0/1/2), ROI x/y/z/w/h/d
|-- projects/knee_OA_dl_app/       reference repo (mafda, cloned)
|-- app/
|   |-- main.py                   Streamlit entry, sidebar + tabs
|   |-- oa_grader.py              Tab1: inference + Grad-CAM
|   |-- mri_viewer.py             Tab2: volume viewer + ROI + JSW proxy
|   |-- implant_sizer.py          Tab3: sizing engine
|   |-- stats_panel.py            Tab4: analytics
|   |-- implant_sizes.json         NexGen dimension tables
`-- scripts/smoke_test.py         verified: Healthy 60% on true KL=0 sample

```

## 5. Six-Hour Execution Timeline (riskiest-first)

| Time   | Phase               | Deliverable / Exit criteria                                                             |
|--------|---------------------|-----------------------------------------------------------------------------------------|
| H0-1   | Setup + risk kill   | [DONE] venv py3.11, TF-CPU 2.15, deps; [DONE] model downloaded; [DONE] smoke test: KL=0 |
| H1-2.5 | Tab 1 OA Grader     | Upload -> grade card + confidence bars + Grad-CAM; sample-image picker                  |
| H2.5-4 | Tab 2 MRI Explorer  | Volume dropdown, slice slider, ROI rectangle, ACL badge, JSW mm readout                 |
| H4-5   | Tab 3 Implant Sizer | Sliders -> ranked NexGen matches + overhang flag; JSON size tables                      |
| H5-5.5 | Tab 4 + polish      | KL/ACL distribution plots; branding; README                                             |
| H5.5-6 | Dry run + buffer    | Full judge walkthrough rehearsed; fallbacks armed                                       |

## 6. Risk Register & Fallbacks

| Risk                             | Mitigation / Fallback                                        |
|----------------------------------|--------------------------------------------------------------|
| TF cannot load legacy HDF5       | [Killed] TF 2.15 + keras 2.15 pinned; load verified          |
| Model download blocked           | [Killed] gdown succeeded (251MB @ ~5MB/s)                    |
| Python 3.14 has no TF wheels     | [Killed] uv-managed Python 3.11.15 venv                      |
| Grad-CAM layer name mismatch     | Resolve layer by type (last GlobalAveragePooling2D input)    |
| Live inference fails during demo | Switch Tab1 to precomputed-results mode (screenshots cached) |
| matplotlib colormap API change   | Use matplotlib.colormaps['jet'] instead of cm.get_cmap       |

## 7. Five-Minute Judge Demo Script

- (0:00) Frame the pain point: meniscal wear assessment and implant sizing today rely on visual inspection and templating - slow, subjective, experience-dependent.
- (0:45) Tab 1 LIVE: upload two contrasting X-rays (grade 0 vs grade 4 from test set) - instant KL grade + confidence + Grad-CAM showing WHERE the model looks (center vs joint edges).
- (2:00) Tab 2: browse a real 3D MRI volume, show ROI box, read out joint-space width in mm; mention metadata carries ACL status enabling population correlation.
- (3:00) Tab 3: slide femoral AP/ML dims -> live ranked implant recommendation with overhang warning - 'this is the Module-2 decision-support story'.

- (4:00) Tab 4 analytics + honest scope slide: what is real AI today, what is proxy, and the GPU-enabled roadmap (nnU-Net on OAI-ZIB/SKM-TEA for true meniscus segmentation).
- (4:40) Close: 'clinician stays in command - we compress minutes of manual assessment to seconds.'

## 8. Post-Hackathon Roadmap (GPU-enabled)

- Train nnU-Net on OAI-ZIB (507 knee MRIs with bone/cartilage/meniscus masks) for true segmentation.
- Replace JSW proxy with genuine meniscus thickness maps at predefined zones (distance-transform on mask).
- Statistical correlation module: thickness vs KL grade / age / sex across cohorts.
- Auto-extract femoral/tibial ML+AP from segmentation instead of manual input; DICOM/PACS ingestion.
- Vendor-neutral sizing: extend implant DB beyond NexGen (Persona, Attune published charts).